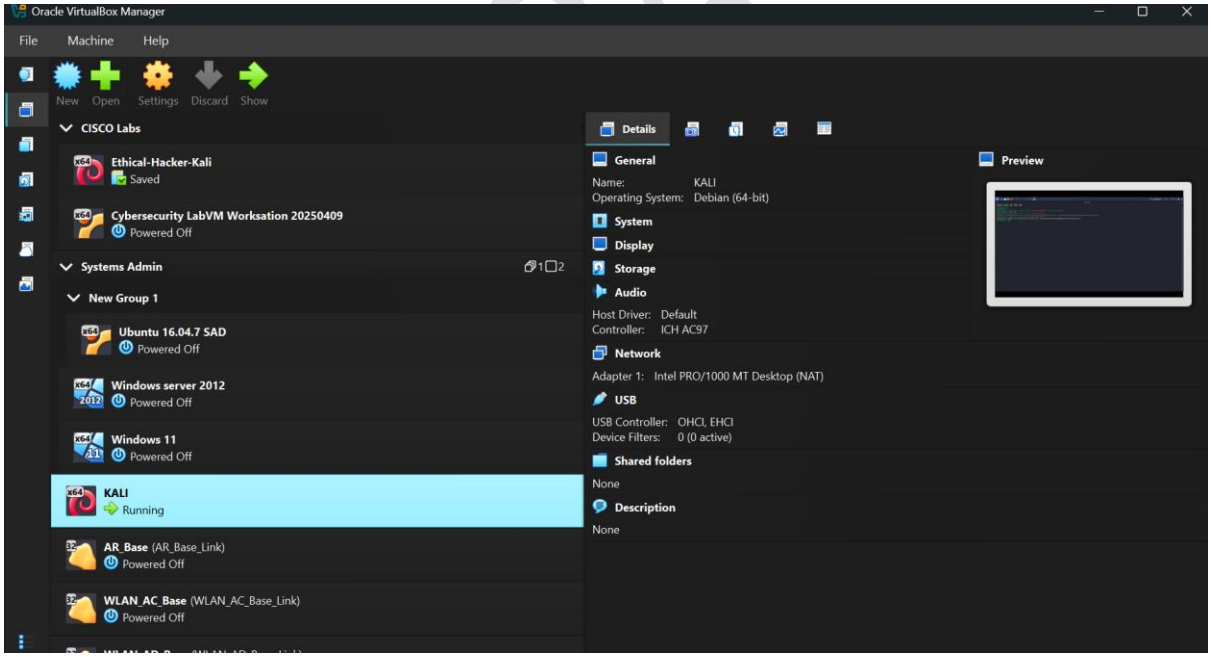
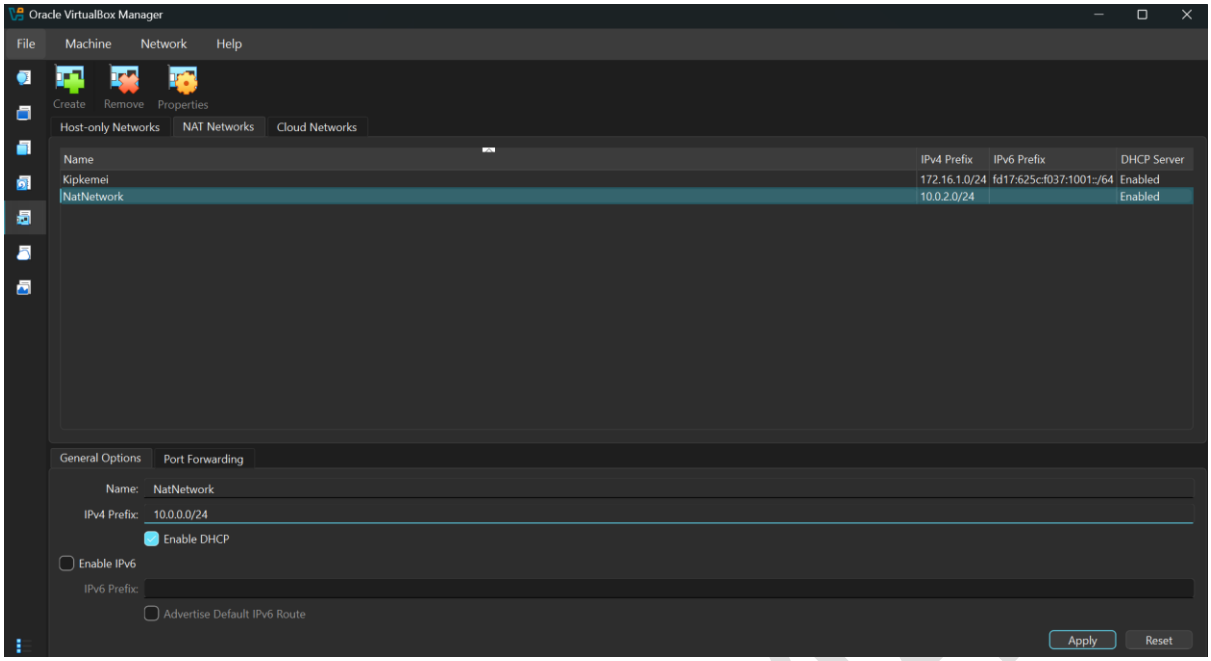
I already had VirtualBox and Kali installed, so no installation was needed. I just verified VirtualBox was updated to the latest version and confirmed 7-Zip was available on my host for handling compressed files.

1. Network configuration
   - Creating the NAT network
     I opened **File > Tools > Network** in VirtualBox, went to the **NAT Networks** tab, and created a new network named NatNetwork. I set the **IPv4 Prefix** to 10.0.0.0/24 and enabled **DHCP** so IPs would be assigned automatically.

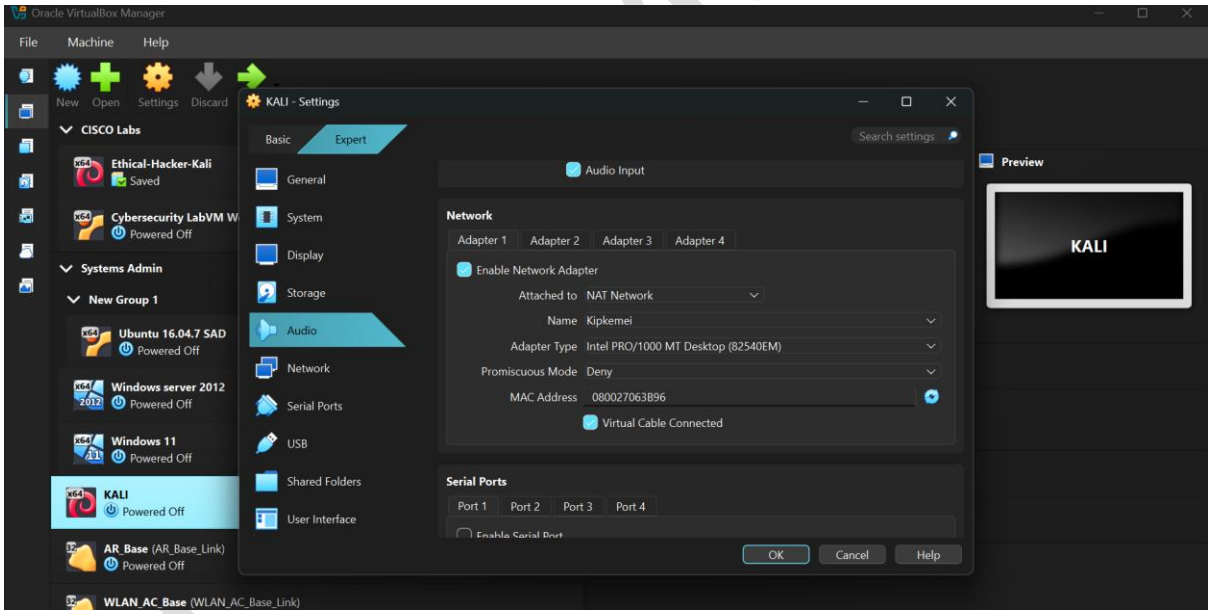
   - Configuring Kali VM Network Settings
     I selected the Kali VM and opened **Settings > Network**. I set **Adapter 1** to attach to **NAT Network** with the name NatNetwork. Under **Advanced**, I set **Promiscuous Mode** to **"Allow All"** so the VM could capture all traffic on the network—essential for cybersecurity testing.

I already had Kali and VirtualBox running, so I didn't need to install, i.e.,

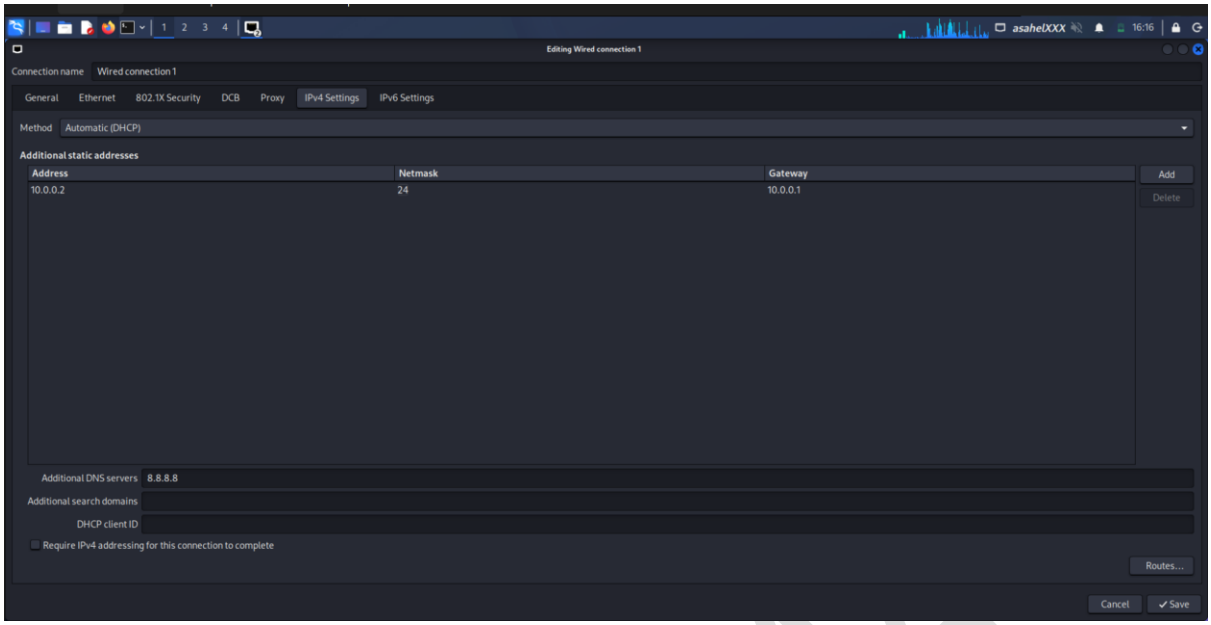




Setting up NAT Network on the network.



Virtualbox Settings for Kali Linux VM

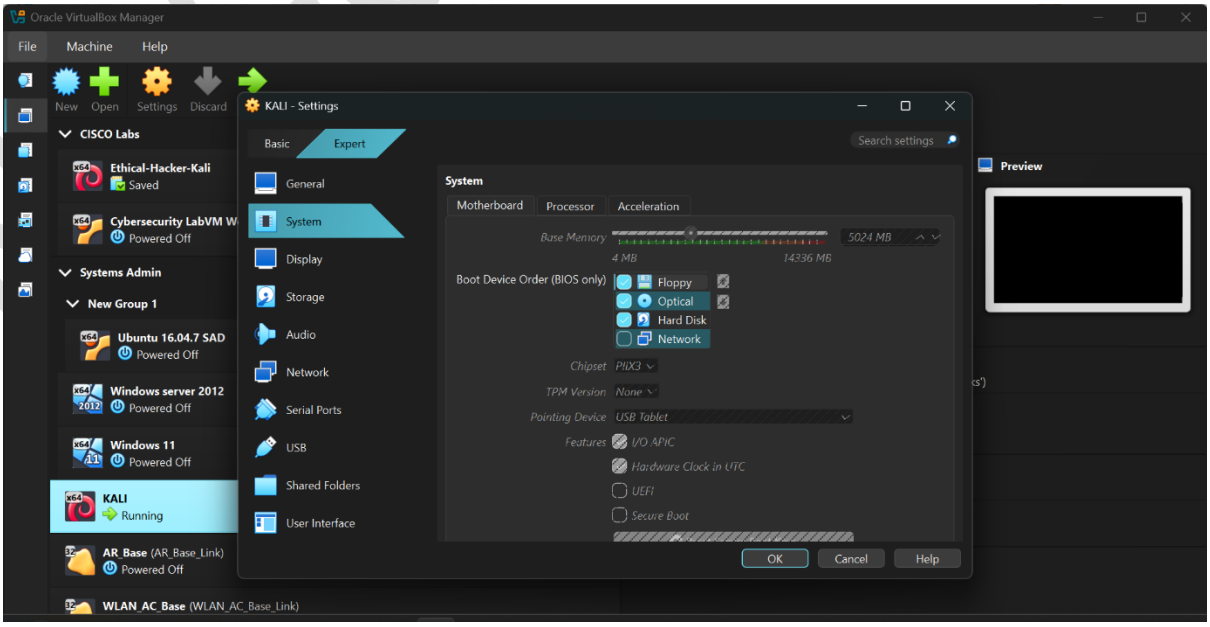


Virtualbox Settings for Kali Linux VM

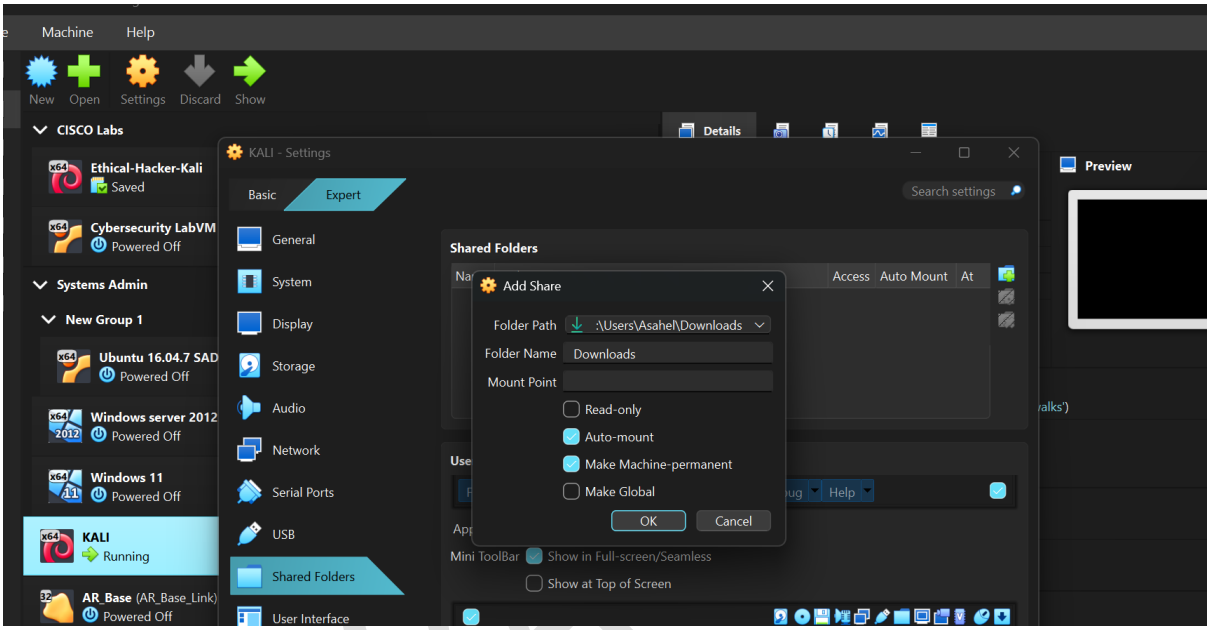
2. VM Hardware setup
   - System and Display Configuration
     To ensure smooth performance, I adjusted the VM resources.

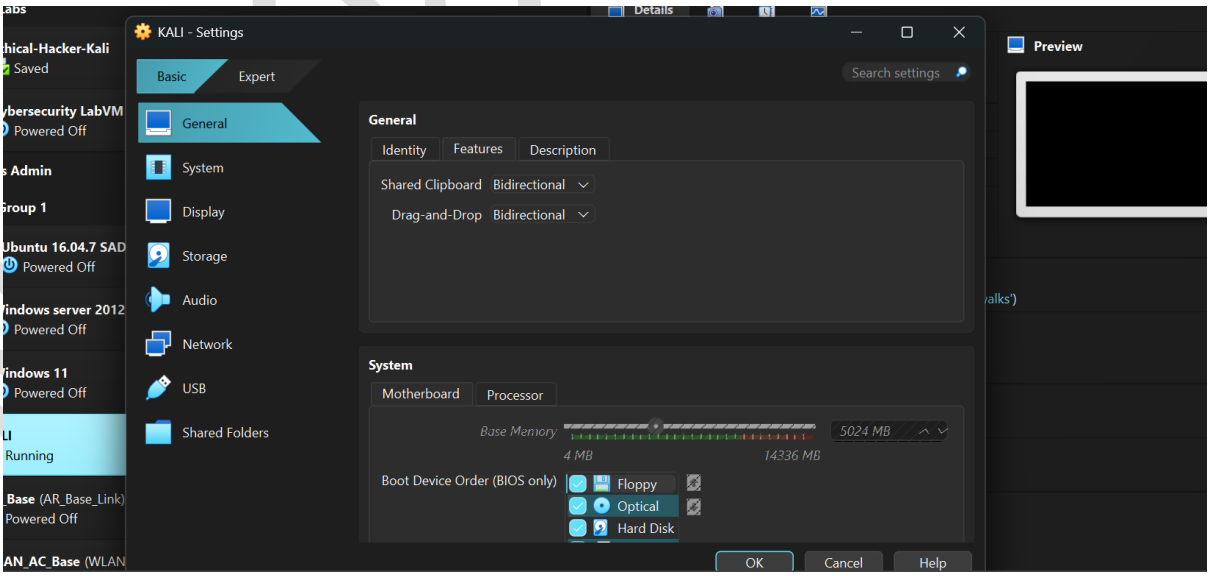
     Under **System > Motherboard**, I allocated sufficient RAM.

     Under **System > Processor**, I assigned multiple CPU cores. I also verified **"Enable Nested Paging"** was checked under **Acceleration** and increased the Video Memory under **Display** for a smoother GUI.



- Enabling Shared Folders and Clipboard
  For easy file transfer between host and VM, I went to **Settings > Shared Folders** and added my host's Downloads folder (C:\Users\user11\Downloads) with **Auto-mount** and **Make Permanent** enabled. I then set both **Shared Clipboard** and **Drag and Drop** to **"Bidirectional"** under **General > Advanced**.



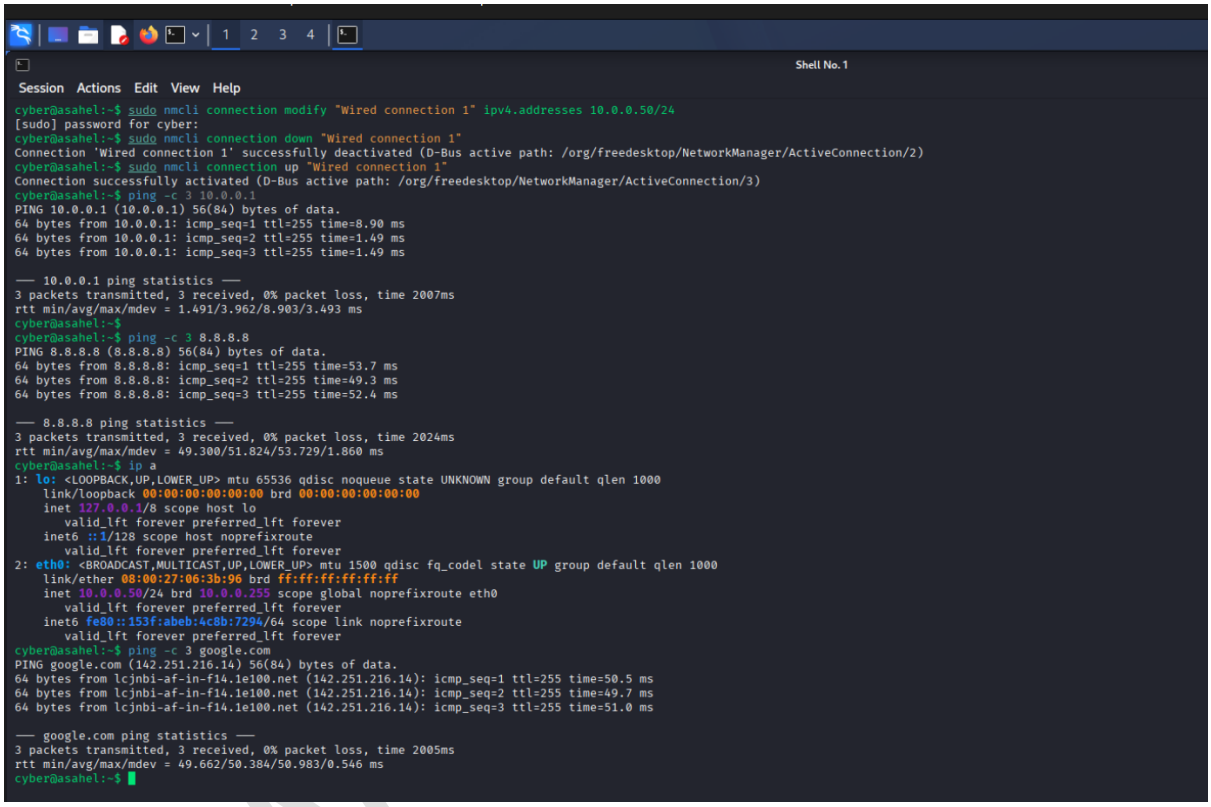


3. IP configuration

Initial Issue: Gateway Unreachable

- After setting a static IP of 10.0.0.2/24, I could not ping the gateway (10.0.0.1) or the internet. The destination was unreachable even though the adapter settings looked correct. This was caused by an address conflict—VirtualBox reserves the lower addresses in a NAT Network for its own gateway and DHCP services, so 10.0.0.2 was not usable.

I reassigned Kali to a higher address within the NAT Network range, i.e., 10.0.0.50 /24



Both the default gateway, Google's DNS, 8.8.8.8 and google.com all return ICMP replies. As seen, the kali machine is indicated as: 10.0.0.50 /24 which falls in the NAT Network range.

Here are the commands:
sudo nmcli connection modify "Wired connection 1" ipv4.addresses 10.0.0.50/24

sudo nmcli connection down "Wired connection 1"

sudo nmcli connection up "Wired connection 1"

- Pinging the **default gateway** (10.0.0.1) returned replies.
- Pinging **Google's DNS** (8.8.8.8) returned replies.
- Pinging **google.com** returned replies, confirming both internet access and DNS resolution were working.



As evidence, this is my Kali, as it was, I just run an update and upgrade.

NW-084-4